# CALCULUS (I)

## EXERSICE SHEET (2)

## DERIVATIVES

(1) For the following functions obtain the tangent and normal lines at given points.

(a) $y = 1+2x-x^3 \qquad (1,2)$

$$m_T = \lim_{h\to 0} \frac{f(1+h)-f(1)}{h} = \lim_{h\to 0} \frac{1+2(1+h)-(1+h)^3-2}{h}$$

$$= \lim_{h\to 0} \frac{\cancel{1}+\cancel{2}+2h-\cancel{1}-3h^2-3h-h^3-\cancel{2}}{h}$$

$$= \lim_{h\to 0} \frac{-h-3h^2-h^3}{h} = \lim_{h\to 0} -1-3h-h^2 = -1$$

$m_T = -1 \qquad m_T \cdot m_N = -1 \qquad m_N = 1$

Tangent line $\quad y-2 = -1(x-1)$

Normal line $\quad y-2 = 1(x-1)$

(b) $y = \sqrt{2x+1} \qquad (4,3)$

$$m_T = \lim_{h\to 0} \frac{f(4+h)-f(4)}{h} = \lim_{h\to 0} \frac{\sqrt{2(4+h)+1}-3}{h}$$

$$= \lim_{h\to 0} \frac{\sqrt{2h+9}-3}{h} \cdot \frac{(\sqrt{2h+9}+3)}{(\sqrt{2h+9}+3)}$$

$$= \lim_{h\to 0} \frac{2\cancel{h}+\cancel{9}-\cancel{9}}{\cancel{h}(\sqrt{2h+9}+3)} = \frac{2}{6} = 1/3$$

$m_N = -3 \qquad$ Tangent line $\quad y-3 = 1/3(x-4)$

Normal line $\quad y-3 = -3(x-4)$

ⓒ $y = \frac{(x-1)}{(x-2)}$ $(3,2)$

$$\lim_{h\to 0} \frac{f(3+h)-f(3)}{h} = \lim_{h\to 0} \frac{\frac{3+h-1}{3+h-2} - 2}{h} = \lim_{h\to 0} \frac{\frac{h+2}{h+1} - 2}{h}$$

$$\lim_{h\to 0} \frac{\frac{h+2}{h+1} - \frac{2(h+1)}{}}{h} = \lim_{h\to 0} \frac{\frac{h+\cancel{2}-2h-\cancel{2}}{h+1}}{h} = \lim_{h\to 0} \frac{-\cancel{h}}{(h+1)\cancel{(h)}} = -1$$

$m_T = -1$ $m_N = 1$

Tangent line $y-2 = -1(x-3)$

Normal line $y-2 = 1(x-3)$

ⓓ $y = 2x/(x+1)^2$ $(0,0)$

$$m_T = \lim_{h\to 0} \frac{f(h)-f(0)}{h} = \lim_{h\to 0} \frac{\frac{2h}{(h+1)^2}}{h} = \lim_{h\to 0} \frac{2}{(h+1)^2} = 2$$

$m_T \cdot m_N = -1$ $m_N = -1/2$

Tangent line $y-0 = 2(x-0)$

Normal line $y-0 = -\frac{1}{2}(x-0)$

② Using formal definition of derivative obtain $f'(a)$ for the following functions

ⓐ $f(x) = 3 - 2x + 4x^2$

$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h} = \lim_{h\to 0} \frac{\cancel{3} - 2(a+h) + 4(a+h)^2 - [\cancel{3} - 2a + 4a^2]}{h}$$

$$= \lim_{h\to 0} \frac{-\cancel{2a} - 2h + \cancel{4a^2} + 8ah + 4h^2 + \cancel{2a} - \cancel{4a^2}}{h}$$

$$= \lim_{h\to 0} -2 + 8a + 4h = -2 + 8a$$

(b) $f(t) = t^4 - 5t$

$$f'(a) = \lim_{z \to a} \frac{f(z) - f(a)}{z - a} = \lim_{z \to a} \frac{(z^4 - 5z) - (a^4 - 5a)}{z - a} = \lim_{z \to a} \frac{(z^4 - a^4) - 5(z-a)}{z - a}$$

$$= \lim_{z \to a} \frac{(z^2 - a^2)(z^2 + a^2) - 5(z-a)}{(z-a)}$$

$$= \lim_{z \to a} \frac{\cancel{(z-a)}(z+a)(z^2+a^2) - 5\cancel{(z-a)}}{\cancel{(z-a)}} = 2a \cdot 2a^2 - 5 = \boxed{4a^3 - 5}$$

(c) $f(x) = \dfrac{2x+1}{x+3}$

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \to 0} \frac{\frac{2a+2h+1}{a+h+3} - \frac{2a+1}{a+3}}{h}$$

$$= \lim_{h \to 0} \frac{\frac{2(a+h)+1}{a+h+3} - \frac{2a+1}{a+3}}{h}$$

$$= \lim_{h \to 0} \frac{(2(a+h)+1)(a+3) - (2a+1)(a+h+3)}{h(a+3)(a+h+3)}$$

$$= \lim_{h \to 0} \frac{(\cancel{2a^2} + \cancel{6a} + \cancel{2ah} + 6h + \cancel{a} + \cancel{3}) - (\cancel{2a^2} + \cancel{2ah} + \cancel{6a} + \cancel{a} + h + \cancel{3})}{h(a+3)(a+h+3)}$$

$$= \lim_{h \to 0} \frac{5\cancel{h}}{\cancel{h}(a+3)(a+h+3)} = \boxed{\frac{5}{(a+3)^2}}$$

(d) $f(x) = \sqrt{3x+1}$

$$f'(a) = \lim_{z \to a} \frac{f(z) - f(a)}{z - a} = \lim_{z \to a} \frac{\sqrt{3z+1} - \sqrt{3a+1}}{z - a} \cdot \frac{(\sqrt{3z+1} + \sqrt{3a+1})}{(\sqrt{3z+1} + \sqrt{3a+1})}$$

$$= \lim_{z \to a} \frac{(3z + \cancel{1}) - (3a + \cancel{1})}{(z-a)(\sqrt{3z+1} + \sqrt{3a+1})} = \lim_{z \to a} \frac{3\cancel{(z-a)}}{\cancel{(z-a)}(\sqrt{3z+1} + \sqrt{3a+1})}$$

$$= \boxed{\frac{3}{2\sqrt{3a+1}}}$$

③ Determine whether $f'(0)$ exist for $f(x) = \begin{cases} x \sin\frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$

$$f'(0) = \lim_{h\to 0} \frac{f(h) - f(0)}{h} = \lim_{h\to 0} \frac{h \cdot \sin 1/h - 0}{h} = \lim_{h\to 0} \sin 1/h$$

Since limit $\sin(1/h)$ doesn't exist $h \to 0$. Therefore $f'(0)$ doesn't exist.
( $\sin(1/h)$ oscillates between $-1$ and $1$ )

④ Find the derivative of the following functions using definition of derivative. find Domain of the function and domain of the derivative function

① $g(x) = \sqrt{1+2x}$ ② $G(t) = \frac{4t}{t+1}$ ③ $f(x) = x + \sqrt{x}$

① $$g'(x) = \lim_{h\to 0} \frac{g(x+h) - g(x)}{h} = \lim_{h\to 0} \frac{\sqrt{1+2(x+h)} - \sqrt{1+2x}}{h}$$

$$= \lim_{h\to 0} \frac{\sqrt{1+2(x+h)} - \sqrt{1+2x}}{h} \cdot \frac{(\sqrt{1+2(x+h)} + \sqrt{1+2x})}{(\sqrt{1+2(x+h)} + \sqrt{1+2x})}$$

$$= \lim_{h\to 0} \frac{\cancel{1} + 2(\cancel{x}+h) - (\cancel{1} + \cancel{2x})}{h(\sqrt{1+2(x+h)} + \sqrt{1+2x})} = \lim_{h\to 0} \frac{\cancel{h}}{\cancel{h}(\sqrt{1+2(x+h)} + \sqrt{1+2x})}$$

$$= \frac{1}{2\sqrt{1+2x}} \qquad D_g = [-1/2, \infty) \qquad g' = (-\tfrac{1}{2}, \infty)$$

② $$G'(t) = \lim_{h\to 0} \frac{G(t+h) - G(t)}{h} = \lim_{h\to 0} \frac{\frac{4(t+h)}{t+h+1} - \frac{4t}{t+1}}{h}$$

$$= \lim_{h\to 0} \frac{4(t+h)(t+1) - 4t(t+h+1)}{h(t+h+1)(t+1)} = \lim_{h\to 0} \frac{4\cancel{(t+h)}[\cancel{(t+1)} - \cancel{t}] - \cancel{4t}}{h(t+h+1)(t+1)}$$

$$= \lim_{h\to 0} \frac{4\cancel{h}}{\cancel{h}(t+h+1)(t+1)} = \frac{4}{(t+1)^2} \qquad D_G = (-\infty, -1) \cup (-1, \infty)$$

$$D_{G'} = (-\infty, -1) \cup (-1, \infty)$$

③ $f'(x) = \lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$

$$= \lim_{h \to 0} \frac{(x+h) + \sqrt{x+h} - x - \sqrt{x}}{h}$$

$$= \lim_{h \to 0} \frac{h + \sqrt{x+h} - \sqrt{x}}{h} = \lim_{h \to 0} 1 + \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

$$= 1 + \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{(\sqrt{x+h} + \sqrt{x})}{\sqrt{x+h} + \sqrt{x}}$$

$$= 1 + \lim_{h \to 0} \frac{(\cancel{x} + \cancel{h}) - \cancel{x}}{\cancel{h}(\sqrt{x+h} + \sqrt{x})} = 1 + \frac{1}{2\sqrt{x}}$$

$D_f = [0, \infty)$ $\quad D_{f'} = (0, \infty)$

## Derivative Rules & Trigonometric Functions

① Obtain the derivatives of the following functions

ⓐ $h(\theta) = \operatorname{cosec}\theta + e^{\theta} \cdot \cot\theta$ $\quad$ ⓑ $y = \dfrac{x}{\cos x}$

ⓒ $y = \dfrac{1 + \sin x}{x + \cos x}$ $\quad$ ⓓ $y = \operatorname{cosec}\theta \, (\theta + \cot\theta)$

ⓐ $h'(\theta) = -\operatorname{cosec}\theta \cdot \cot\theta + e^{\theta}\cot\theta - e^{\theta}\operatorname{cosec}^2\theta$

ⓑ $y = \dfrac{x}{\cos x}$ $\quad y' = \dfrac{\cos x + \sin x \cdot x}{\cos^2 x}$

ⓒ $y' = \dfrac{\cos x\,(x + \cos x) - (1 - \sin x)(1 + \sin x)}{(x + \cos x)^2}$

$$= \frac{\cos x \cdot x + \cos^2 x - \underbrace{(1 - \sin^2 x)}_{\cos^2 x}}{(x + \cos x)^2} = \frac{x\cos x}{(x + \cos x)^2}.$$

(2) Find an equation of the tangent line to the curve at given point

ⓐ $y = \tan x$ $(\pi/4, 1)$ ⓑ $y = e^x \cos x$ $(0,1)$

ⓒ $y = x + \cos x$ $(0,1)$ ⓓ $y = \dfrac{1}{\sin x + \cos x}$ $(0,1)$

ⓐ $y' = \sec^2 x$ $m_T\Big|_{(\pi/4, 1)} = \sec^2 \pi/4 = 2$

$y - 1 = 2(x - \pi/4)$

ⓑ $y' = e^x \cos x - \sin x . e^x$ $m_T\Big|_{(0,1)} = e^0 . \cos 0 - \sin 0 . e^0 = 1$

$y - 1 = 1(x - 0)$ $\Rightarrow$ $y = x + 1$

ⓒ $y = x + \cos x$ $(0,1)$ $y - 1 = 1(x - 0)$

$y' = 1 - \sin x$ $m_T\Big|_{(0,1)} = 1 - \sin 0 = 1$

ⓓ $y = \dfrac{1}{\sin x + \cos x}$ $y' = \dfrac{(\cos x - \sin x)}{(\sin x + \cos x)^2}$

$m_T\Big|_{(0,1)} = \left(\dfrac{\cos 0 - \sin 0}{(\sin 0 + \cos 0)^2}\right) = 1$ $y - 1 = 1(x - 0)$

(3) Evaluate the following limits.

ⓐ $\lim\limits_{x \to 0} \dfrac{\sin 4x}{\sin 6x}$ ⓑ $\lim\limits_{t \to 0} \dfrac{\tan 6t}{\sin 2t}$ ⓒ $\lim\limits_{\theta \to 0} \dfrac{\cos \theta - 1}{\sin \theta}$

ⓓ $\lim\limits_{\theta \to 0} \dfrac{\sin(\cos \theta)}{\sec \theta}$

(a) $\lim_{x\to 0} \frac{\sin 4x}{\sin 6x} = \lim_{x\to 0} \sin 4x \cdot \frac{4x}{4x} \cdot \frac{1}{\sin 6x} \frac{6x}{6x}$

$= \lim_{x\to 0} \frac{\sin 4x}{4x} \cdot \frac{4x}{6x} \cdot \frac{6x}{\sin 6x} = \frac{4}{6} \lim_{x\to 0} \frac{\sin 4x}{4x} \cdot \lim_{x\to 0} \frac{6x}{\sin 6x}$

$x \to 0 \quad u = 4x \to 0.$ $\qquad \lim_{x\to 0} \frac{\sin 4x}{4x} = \lim_{u\to 0} \frac{\sin u}{u} = 1$

$x \to 0 \quad v = 6x \to 0.$ $\qquad \lim_{x\to 0} 1/\frac{\sin 6x}{6x} = \lim_{v\to 0} 1/\frac{\sin v}{v} = 1.$

$= \frac{4}{6} \left(\lim_{x\to 0} \frac{\sin 4x}{4x}\right)^1 \cdot \lim_{x\to 0} \left(\frac{6x}{\sin 6x}\right) = 1.$

(b) $\lim_{t\to 0} \frac{\tan 6t}{\sin 2t} = \lim_{t\to 0} \frac{\sin 6t}{\cos 6t} \cdot \frac{1}{\sin 2t} \cdot \frac{6t}{6t}.$

$3.2t.$

$= \lim_{t\to 0} \frac{\sin 6t}{6t} \cdot \frac{6t}{\sin 2t} \cdot \frac{1}{\cos 6t} = \boxed{3}$

$\lim_{t\to 0} \frac{\sin 6t}{6t} = \lim_{u\to 0.} \frac{\sin u}{u} = 1.$

$6t = u \quad t \to 0 \quad u \to 0$

$\lim_{t\to 0} \frac{2t}{\sin 2t} = \lim_{t\to 0} \frac{1}{\frac{\sin 2t}{2t}} = \lim_{v\to 0.} \frac{1}{\frac{\sin v}{v}} = 1$

$t \to 0 \quad v = 2t \to 0$

$\lim_{t\to 0} \frac{1}{\cos 6t} = 1.$

(c) $\lim_{\theta\to 0} \frac{\cos\theta - 1}{\sin\theta} = \lim_{\theta\to 0.} \frac{\cos\theta - 1}{\sin\theta} \cdot \frac{\theta}{\theta} = \lim_{\theta\to 0} \frac{\frac{\cos\theta - 1}{\theta}}{\frac{\sin\theta}{\theta}}.$

$= \frac{\left(\lim_{\theta\to 0} \frac{\cos\theta - 1}{\theta}\right)^0}{\left(\lim_{\theta\to 0} \frac{\sin\theta}{\theta}\right) \to 1} = 0.$

$\lim_{\theta\to 0} \frac{\cos\theta - 1}{\theta} \left(\frac{\cos\theta + 1}{\cos\theta + 1}\right.$

$= \lim_{\theta\to 0} \frac{-\sin^2\theta}{\theta(\cos\theta + 1)}$

$= \lim \left(\frac{\sin\theta}{\ }\right) \left(-\frac{\sin\theta}{\ }\right) =$

(d) $\lim_{\theta \to 0} \frac{\sin(\cos\theta)}{\sec\theta} = \frac{\sin(\lim_{\theta \to 0} \cos\theta)}{\lim_{\theta \to 0} \sec\theta} = \frac{\sin 1}{1} = \sin 1$

## CHAIN RULE

TO BE CONTINUED...

# The Chain Rule

**Question 1:** State the Chain Rule using Leibniz notation.
**Solution:** If $y = f(u)$ and $u = t(x)$ are both differentiable functions, the Chain Rule is

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}.$$

—

**Question 2:** Differentiate $y = (x^3 - 1)^{100}$ with respect to $x$.
**Solution:** We treat this as $y = u^{100}$ where $u = x^3 - 1$.

$$\frac{dy}{dx} = 100(x^3 - 1)^{99} \cdot 3x^2 = 300x^2(x^3 - 1)^{99}.$$

—

**Question 3:** Find the derivative $F'(x)$ for $F(x) = \sqrt{x^2 + 1}$.
**Solution:** Writing $F(x) = (x^2 + 1)^{1/2}$,

$$F'(x) = \frac{1}{2}(x^2 + 1)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2 + 1}}.$$

—

**Question 4:** Find the derivative of $y = e^{\sin x}$.
**Solution:** Let $u = \sin x$.

$$\frac{dy}{dx} = e^{\sin x} \cos x.$$

—

**Question 5:** What is the differentiation formula for $a^x$, where $a > 0$?
**Solution:**

$$\frac{d}{dx}(a^x) = a^x \ln a.$$

—

**Question 6:** Differentiate $y = \sin^2 x$.
**Solution:**

$$\frac{dy}{dx} = 2 \sin x \cos x.$$

—

**Question 7:** Find the derivative of $y = \ln(x^3 + 1)$.
**Solution:**

$$\frac{dy}{dx} = \frac{1}{x^3 + 1} \cdot 3x^2 = \frac{3x^2}{x^3 + 1}.$$

—

**Question 8:** Find the derivative $F'(x)$ for $F(x) = \sqrt{x^2 + x + 1}$.
**Solution:**

$$F'(x) = \frac{1}{2}(x^2 + x + 1)^{-1/2}(2x + 1) = \frac{2x + 1}{2\sqrt{x^2 + x + 1}}.$$

—

**Question 9:** Find the derivative of $y = \ln |x|$.
**Solution:**

$$\frac{dy}{dx} = \frac{1}{x}.$$

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**Question 10:** Find the derivative of $y = \sec x$.
**Solution:**

$$\frac{dy}{dx} = \sec x \tan x.$$

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**Question 11:** Find the derivative of $y = \csc x$.
**Solution:**

$$\frac{dy}{dx} = -\csc x \cot x.$$

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**Question 12:** Find the derivative of $y = \cot x$.
**Solution:**

$$\frac{dy}{dx} = -\csc^2 x.$$

—

**Question 13:** Differentiate $y = \sin(\cos(\tan x))$.
**Solution:**

$$y' = \cos(\cos(\tan x)) \cdot [-\sin(\tan x)] \cdot \sec^2 x = -\cos(\cos(\tan x))\sin(\tan x)\sec^2 x.$$

—

**Question 14:** Differentiate $y = e^{\sec(3\theta)}$ with respect to $\theta$.
**Solution:**

$$\frac{dy}{d\theta} = e^{\sec(3\theta)}\sec(3\theta)\tan(3\theta)\cdot 3 = 3e^{\sec(3\theta)}\sec(3\theta)\tan(3\theta).$$

—

**Question 15:** Find the derivative of $y = \dfrac{r}{\sqrt{r^2+1}}$ with respect to $r$.
**Solution:**

$$\frac{dy}{dr} = (r^2+1)^{-1/2} - \frac{r^2}{(r^2+1)^{3/2}} = \frac{1}{(r^2+1)^{3/2}}.$$

—

**Question 16:** Find the derivative of $y = \cot^2(\sin x)$.
**Solution:**

$$\frac{dy}{dx} = 2\cot(\sin x)\cdot[-\csc^2(\sin x)]\cdot\cos x = -2\cos x\cot(\sin x)\csc^2(\sin x).$$

—

**Question 17:** Differentiate $y = e^{\cos x} + \cos(e^x)$.
**Solution:**

$$\frac{dy}{dx} = -\sin x\, e^{\cos x} - e^x\sin(e^x).$$

—

**Question 18:** Find $\dfrac{dy}{dx}$ for $y = e^{-5x}\cos(3x)$.
**Solution:**

$$\frac{dy}{dx} = -5e^{-5x}\cos 3x - 3e^{-5x}\sin 3x = -e^{-5x}(5\cos 3x + 3\sin 3x).$$

—

**Question 19:** Find the derivative of $y = \sqrt{x+\sqrt{x}}$.
**Solution:**

$$y = (x + x^{1/2})^{1/2}, \quad \frac{dy}{dx} = \frac{1}{2\sqrt{x+\sqrt{x}}}\left(1 + \frac{1}{2\sqrt{x}}\right).$$

—

**Question 20:** Differentiate $y = \ln|\csc 5x|$.
**Solution:**

$$\frac{dy}{dx} = \frac{1}{\csc 5x}(-5\csc 5x\cot 5x) = -5\cot 5x.$$

## Implicit Differentiation

**Question 1:** Find $\frac{dy}{dx}$ for the equation $x^2 + y^2 = 36$.
**Solution:**

$$2x + 2y\frac{dy}{dx} = 0 \quad \Rightarrow \quad \frac{dy}{dx} = -\frac{x}{y}.$$

—

**Question 2:** Find $\frac{dy}{dx}$ for $x^3 + y^3 = 6xy$.
**Solution:**

$$3x^2 + 3y^2\frac{dy}{dx} = 6y + 6x\frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx}(3y^2 - 6x) = 6y - 3x^2 \Rightarrow \frac{dy}{dx} = \frac{2y - x^2}{y^2 - 2x}.$$

—

**Question 3:** Find $\frac{dy}{dx}$ for $x^2 + xy^2 = 4$.
**Solution:**

$$2x + y^2 + 2xy\frac{dy}{dx} = 0 \Rightarrow 2xy\frac{dy}{dx} = -2x - y^2 \Rightarrow \frac{dy}{dx} = -\frac{2x + y^2}{2xy}.$$

—

**Question 4:** Find $\frac{dy}{dx}$ for $x^2y + xy^2 = 3x$.
**Solution:**

$$(2xy + x^2\frac{dy}{dx}) + (y^2 + 2xy\frac{dy}{dx}) = 3$$

$$\Rightarrow \frac{dy}{dx}(x^2 + 2xy) = 3 - 2xy - y^2 \Rightarrow \frac{dy}{dx} = \frac{3 - 2xy - y^2}{x^2 + 2xy}.$$

—

**Question 5:** Find $\frac{dy}{dx}$ for $\tan(x - y) = y$.
**Solution:**

$$\sec^2(x-y)\left(1 - \frac{dy}{dx}\right) = \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{\sec^2(x-y)}{1 + \sec^2(x-y)}.$$

—

**Question 6:** Find the slope of the tangent line to $x^2 - xy + y^2 = 3$ at $(1, 1)$.
**Solution:**

$$2x - y - x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0.$$

At $(1, 1)$ this becomes

$$2 - 1 - \frac{dy}{dx} + 2\frac{dy}{dx} = 0 \Rightarrow 1 + \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -1.$$

—

**Question 7:** Derive the formula for $\frac{d}{dx}(\sin^{-1} x)$.
**Solution:** Let $x = \sin y$. Then

$$1 = \cos y\frac{dy}{dx} \quad \Rightarrow \quad \frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1 - x^2}}.$$

—

**Question 8:** Find $\frac{dy}{dx}$ for $e^{xy} = x + y$.
**Solution:**

$$e^{xy}(y + x\frac{dy}{dx}) = 1 + \frac{dy}{dx} \Rightarrow \frac{dy}{dx}(xe^{xy} - 1) = 1 - ye^{xy} \Rightarrow \frac{dy}{dx} = \frac{1 - ye^{xy}}{xe^{xy} - 1}.$$

—

**Question 9:** Find $\frac{dx}{dy}$ for $x^2y + xy^2 = 3x$, treating $y$ as the independent variable.

**Solution:** Differentiating with respect to $y$:

$$(2x\frac{dx}{dy}y + x^2) + (y^2\frac{dx}{dy} + 2xy) = 3\frac{dx}{dy}$$

$$\Rightarrow \frac{dx}{dy}(2xy + y^2 - 3) = -x^2 - 2xy \Rightarrow \frac{dx}{dy} = \frac{-x^2 - 2xy}{2xy + y^2 - 3}.$$

—

**Question 10:** Find the slope of the tangent line to $x^2 + 4y^2 = 36$ at $x = 0$.
**Solution:**

$$2x + 8y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{4y}.$$

At $x = 0$ this is 0.

—

**Question 11:** Find $\frac{dy}{dx}$ for $y \sin x = x \cos y$.
**Solution:**

$$\frac{dy}{dx}\sin x + y\cos x = \cos y - x\sin y\frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx}(\sin x + x\sin y) = \cos y - y\cos x \Rightarrow \frac{dy}{dx} = \frac{\cos y - y\cos x}{\sin x + x\sin y}.$$

—

**Question 12:** Derive the formula for $\frac{d}{dx}(\cos^{-1} x)$.
**Solution:** Let $x = \cos y$.

$$1 = -\sin y\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = -\frac{1}{\sin y} = -\frac{1}{\sqrt{1 - x^2}}.$$

—

**Question 13:** Find $f'(1)$ if $1 + f(x) + x^2[f(x)]^3 = 0$ and $f(1) = 2$.
**Solution:** Differentiate:

$$f'(x) + 2x[f(x)]^3 + x^2 \cdot 3[f(x)]^2 f'(x) = 0.$$

At $x = 1$, $f(1) = 2$:

$$f'(1) + 2 \cdot 1 \cdot 8 + 1^2 \cdot 3 \cdot 4\, f'(1) = 0 \Rightarrow f'(1) + 16 + 12f'(1) = 0 \Rightarrow f'(1) = -\frac{16}{13}.$$

—

**Question 14:** What is the formula for $\frac{d}{dx}(\csc^{-1} x)$?
**Solution:**

$$\frac{d}{dx}(\csc^{-1} x) = -\frac{1}{x\sqrt{x^2 - 1}}.$$

—

**Question 15:** For $x^3y^2$, explain how the Chain Rule is used in implicit differentiation.
**Solution:** Since $y = y(x)$,

$$\frac{d}{dx}(y^2) = 2y\frac{dy}{dx},$$

so

$$\frac{d}{dx}(x^3y^2) = 3x^2y^2 + x^3 \cdot 2y\frac{dy}{dx}.$$

—

**Question 16:** Find $\frac{dy}{dx}$ for $xe^y = y + 1$.
**Solution:**

$$e^y + xe^y\frac{dy}{dx} = \frac{dy}{dx} \Rightarrow \frac{dy}{dx}(xe^y - 1) = -e^y \Rightarrow \frac{dy}{dx} = \frac{-e^y}{xe^y - 1}.$$

—

**Question 17:** Find $\frac{dy}{dx}$ for $\sin(xy) = x^2 + y$.

**Solution:**

$$\cos(xy)(y + x\frac{dy}{dx}) = 2x + \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx}(x\cos(xy) - 1) = 2x - y\cos(xy) \Rightarrow \frac{dy}{dx} = \frac{2x - y\cos(xy)}{x\cos(xy) - 1}.$$

—

**Question 18:** Find $\frac{dy}{dx}$ for $x^2 \cos y + \sin(2y) = xy$.

**Solution:**

$$2x\cos y - x^2 \sin y\frac{dy}{dx} + 2\cos(2y)\frac{dy}{dx} = y + x\frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx}(2\cos(2y) - x^2 \sin y - x) = y - 2x\cos y \Rightarrow \frac{dy}{dx} = \frac{y - 2x\cos y}{2\cos(2y) - x^2 \sin y - x}.$$

—

**Question 19:** Find $\frac{dy}{dx}$ for $xy^4 + x^2 y = x + 3y$.

**Solution:**

$$(y^4 + 4xy^3\frac{dy}{dx}) + (2xy + x^2\frac{dy}{dx}) = 1 + 3\frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx}(4xy^3 + x^2 - 3) = 1 - y^4 - 2xy \Rightarrow \frac{dy}{dx} = \frac{1 - y^4 - 2xy}{4xy^3 + x^2 - 3}.$$

—

**Question 20:** Find the points on $4x^2 + 9y^2 = 36$ where the tangent line is vertical.

**Solution:**

$$8x + 18y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{8x}{18y}.$$

Vertical tangents occur when the denominator is 0, so $y = 0$. Then $4x^2 = 36 \Rightarrow x = \pm 3$. The points are $(3, 0)$ and $(-3, 0)$.

## Inverse Function Derivative Solutions

**Question 1:** $f(x) = 2x + 1$. Find $(f^{-1})'(5)$.
**Solution Steps**

1. Find $x_0$ such that $f(x_0) = 5$: $2x_0 + 1 = 5 \implies x_0 = 2$. So $f^{-1}(5) = 2$.
2. Find $f'(x)$: $f'(x) = 2$.
3. Evaluate $f'(x_0)$: $f'(2) = 2$.
4. Apply the theorem: $(f^{-1})'(5) = \frac{1}{f'(2)}$.

$$(f^{-1})'(5) = \frac{1}{2}$$

—
**Question 2:** $f(x) = x^3 + 1$. Find $(f^{-1})'(9)$.
**Solution Steps**

1. Find $x_0$: $x_0^3 + 1 = 9 \implies x_0^3 = 8 \implies x_0 = 2$. So $f^{-1}(9) = 2$.
2. Find $f'(x)$: $f'(x) = 3x^2$.
3. Evaluate $f'(x_0)$: $f'(2) = 3(2)^2 = 12$.
4. Apply the theorem: $(f^{-1})'(9) = \frac{1}{f'(2)}$.

$$(f^{-1})'(9) = \frac{1}{12}$$

—
**Question 3:** $f(x) = x^5 - 3x$. Find $(f^{-1})'(-2)$.
**Solution Steps**

1. Find $x_0$: $x_0^5 - 3x_0 = -2 \implies x_0 = 1$ is a solution $(1 - 3 = -2)$. So $f^{-1}(-2) = 1$.
2. Find $f'(x)$: $f'(x) = 5x^4 - 3$.
3. Evaluate $f'(x_0)$: $f'(1) = 5(1)^4 - 3 = 2$.
4. Apply the theorem: $(f^{-1})'(-2) = \frac{1}{f'(1)}$.

$$(f^{-1})'(-2) = \frac{1}{2}$$

—
**Question 4:** $f(x) = \frac{1}{2}x^3 + x$. Find $(f^{-1})'(6)$.
**Solution Steps**

1. Find $x_0$: $\frac{1}{2}x_0^3 + x_0 = 6$. Testing $x_0 = 2$: $\frac{1}{2}(8) + 2 = 4 + 2 = 6$. So $f^{-1}(6) = 2$.
2. Find $f'(x)$: $f'(x) = \frac{3}{2}x^2 + 1$.
3. Evaluate $f'(x_0)$: $f'(2) = \frac{3}{2}(2)^2 + 1 = 6 + 1 = 7$.
4. Apply the theorem: $(f^{-1})'(6) = \frac{1}{f'(2)}$.

$$(f^{-1})'(6) = \frac{1}{7}$$

—
**Question 5:** $f(x) = e^{2x} + x$. Find $(f^{-1})'(1)$.
**Solution Steps**

1. Find $x_0$: $e^{2x_0} + x_0 = 1$. Testing $x_0 = 0$: $e^0 + 0 = 1$. So $f^{-1}(1) = 0$.
2. Find $f'(x)$: $f'(x) = 2e^{2x} + 1$.

3. Evaluate $f'(x_0)$: $f'(0) = 2e^0 + 1 = 2 + 1 = 3$.
4. Apply the theorem: $(f^{-1})'(1) = \frac{1}{f'(0)}$.

$$(f^{-1})'(1) = \frac{1}{3}$$

—
**Question 6:** $f(x) = \ln(x) + x$. Find $(f^{-1})'(1)$.
**Solution Steps**

1. Find $x_0$: $\ln(x_0) + x_0 = 1$. Testing $x_0 = 1$: $\ln(1) + 1 = 0 + 1 = 1$. So $f^{-1}(1) = 1$.
2. Find $f'(x)$: $f'(x) = \frac{1}{x} + 1$.
3. Evaluate $f'(x_0)$: $f'(1) = \frac{1}{1} + 1 = 2$.
4. Apply the theorem: $(f^{-1})'(1) = \frac{1}{f'(1)}$.

$$(f^{-1})'(1) = \frac{1}{2}$$

—
**Question 7:** $f(x) = x^5 + 2x^3$. Find $(f^{-1})'(3)$.
**Solution Steps**

1. Find $x_0$: $x_0^5 + 2x_0^3 = 3$. Testing $x_0 = 1$: $1 + 2 = 3$. So $f^{-1}(3) = 1$.
2. Find $f'(x)$: $f'(x) = 5x^4 + 6x^2$.
3. Evaluate $f'(x_0)$: $f'(1) = 5(1) + 6(1) = 11$.
4. Apply the theorem: $(f^{-1})'(3) = \frac{1}{f'(1)}$.

$$(f^{-1})'(3) = \frac{1}{11}$$

—
**Question 8:** $f(x) = x^3 + 3x^2 + 4x$. Find $(f^{-1})'(8)$.
**Solution Steps**

1. Find $x_0$: $x_0^3 + 3x_0^2 + 4x_0 = 8$. Testing $x_0 = 1$: $1 + 3 + 4 = 8$. So $f^{-1}(8) = 1$.
2. Find $f'(x)$: $f'(x) = 3x^2 + 6x + 4$.
3. Evaluate $f'(x_0)$: $f'(1) = 3(1) + 6(1) + 4 = 13$.
4. Apply the theorem: $(f^{-1})'(8) = \frac{1}{f'(1)}$.

$$(f^{-1})'(8) = \frac{1}{13}$$

—
**Question 9:** $f(x) = 4x - x^3$. Find $(f^{-1})'(3)$.
**Solution Steps**

1. Find $x_0$: $4x_0 - x_0^3 = 3$. Testing $x_0 = 1$: $4 - 1 = 3$. So $f^{-1}(3) = 1$.
2. Find $f'(x)$: $f'(x) = 4 - 3x^2$.
3. Evaluate $f'(x_0)$: $f'(1) = 4 - 3(1) = 1$.
4. Apply the theorem: $(f^{-1})'(3) = \frac{1}{f'(1)}$.

$$(f^{-1})'(3) = 1$$

—
**Question 10:** $f(x) = \frac{x-1}{x-2}$. Find $(f^{-1})'(0)$.
**Solution Steps**

1. Find $x_0$: $\frac{x_0-1}{x_0-2} = 0 \implies x_0 - 1 = 0 \implies x_0 = 1$. So $f^{-1}(0) = 1$.

2. Find $f'(x)$: $f'(x) = \frac{(1)(x-2)-(x-1)(1)}{(x-2)^2} = \frac{-1}{(x-2)^2}$.

3. Evaluate $f'(x_0)$: $f'(1) = \frac{-1}{(1-2)^2} = -1$.

4. Apply the theorem: $(f^{-1})'(0) = \frac{1}{f'(1)}$.

$$(f^{-1})'(0) = -1$$

—

**Question 11:** $f(x) = 5x^3 + 3x$. Find $(f^{-1})'(8)$.
**Solution Steps**

1. Find $x_0$: $5x_0^3 + 3x_0 = 8$. Testing $x_0 = 1$: $5 + 3 = 8$. So $f^{-1}(8) = 1$.

2. Find $f'(x)$: $f'(x) = 15x^2 + 3$.

3. Evaluate $f'(x_0)$: $f'(1) = 15(1) + 3 = 18$.

4. Apply the theorem: $(f^{-1})'(8) = \frac{1}{f'(1)}$.

$$(f^{-1})'(8) = \frac{1}{18}$$

—

**Question 12:** $f(x) = 2\sin x + x$. Find $(f^{-1})'(0)$.
**Solution Steps**

1. Find $x_0$: $2\sin x_0 + x_0 = 0$. Testing $x_0 = 0$: $2\sin(0) + 0 = 0$. So $f^{-1}(0) = 0$.

2. Find $f'(x)$: $f'(x) = 2\cos x + 1$.

3. Evaluate $f'(x_0)$: $f'(0) = 2\cos(0) + 1 = 2(1) + 1 = 3$.

4. Apply the theorem: $(f^{-1})'(0) = \frac{1}{f'(0)}$.

$$(f^{-1})'(0) = \frac{1}{3}$$

—

**Question 13:** $f(x) = 2e^x - e^{-x}$. Find $(f^{-1})'(1)$.
**Solution Steps**

1. Find $x_0$: $2e^{x_0} - e^{-x_0} = 1$. Testing $x_0 = 0$: $2(1) - 1 = 1$. So $f^{-1}(1) = 0$.

2. Find $f'(x)$: $f'(x) = 2e^x + e^{-x}$.

3. Evaluate $f'(x_0)$: $f'(0) = 2e^0 + e^0 = 2 + 1 = 3$.

4. Apply the theorem: $(f^{-1})'(1) = \frac{1}{f'(0)}$.

$$(f^{-1})'(1) = \frac{1}{3}$$

—

**Question 14:** $f(x) = x^7 + 5x$. Find $(f^{-1})'(-6)$.
**Solution Steps**

1. Find $x_0$: $x_0^7 + 5x_0 = -6$. Testing $x_0 = -1$: $(-1)^7 + 5(-1) = -1 - 5 = -6$. So $f^{-1}(-6) = -1$.

2. Find $f'(x)$: $f'(x) = 7x^6 + 5$.

3. Evaluate $f'(x_0)$: $f'(-1) = 7(-1)^6 + 5 = 7 + 5 = 12$.

4. Apply the theorem: $(f^{-1})'(-6) = \frac{1}{f'(-1)}$.

$$(f^{-1})'(-6) = \frac{1}{12}$$

—

**Question 15:** $f(x) = \tan x + x$. Find $(f^{-1})'(\pi/4 + 1)$.
**Solution Steps**

1. Find $x_0$: $\tan x_0 + x_0 = \pi/4 + 1$. Testing $x_0 = \pi/4$: $\tan(\pi/4) + \pi/4 = 1 + \pi/4$. So $f^{-1}(\pi/4+1) = \pi/4$.
2. Find $f'(x)$: $f'(x) = \sec^2 x + 1$.
3. Evaluate $f'(x_0)$: $f'(\pi/4) = \sec^2(\pi/4) + 1 = 2 + 1 = 3$.
4. Apply the theorem: $(f^{-1})'(\pi/4+1) = \frac{1}{f'(\pi/4)}$.

$$(f^{-1})'(\pi/4+1) = \frac{1}{3}$$

—

**Question 16:** $f(x) = \frac{x+1}{x}$. Find $(f^{-1})'(3)$.
**Solution Steps**

1. Find $x_0$: $\frac{x_0+1}{x_0} = 3 \implies x_0 + 1 = 3x_0 \implies 1 = 2x_0 \implies x_0 = 1/2$. So $f^{-1}(3) = 1/2$.
2. Rewrite $f(x) = 1 + x^{-1}$, so $f'(x) = -x^{-2} = -\frac{1}{x^2}$.
3. Evaluate $f'(x_0)$: $f'(1/2) = -\frac{1}{(1/2)^2} = -4$.
4. Apply the theorem: $(f^{-1})'(3) = \frac{1}{f'(1/2)}$.

$$(f^{-1})'(3) = -\frac{1}{4}$$

—

**Question 17:** $f(x) = x^3 + 5x^2$. Find $(f^{-1})'(6)$.
**Solution Steps**

1. Find $x_0$: $x_0^3 + 5x_0^2 = 6$. Testing $x_0 = 1$: $1 + 5 = 6$. So $f^{-1}(6) = 1$.
2. Find $f'(x)$: $f'(x) = 3x^2 + 10x$.
3. Evaluate $f'(x_0)$: $f'(1) = 3(1) + 10(1) = 13$.
4. Apply the theorem: $(f^{-1})'(6) = \frac{1}{f'(1)}$.

$$(f^{-1})'(6) = \frac{1}{13}$$

—

**Question 18:** $f(x) = x^3 + x$. Find $(f^{-1})'(10)$.
**Solution Steps**

1. Find $x_0$: $x_0^3 + x_0 = 10$. Testing $x_0 = 2$: $8 + 2 = 10$. So $f^{-1}(10) = 2$.
2. Find $f'(x)$: $f'(x) = 3x^2 + 1$.
3. Evaluate $f'(x_0)$: $f'(2) = 3(2)^2 + 1 = 12 + 1 = 13$.
4. Apply the theorem: $(f^{-1})'(10) = \frac{1}{f'(2)}$.

$$(f^{-1})'(10) = \frac{1}{13}$$

—

**Question 19:** $f(x) = \sqrt{x+3} + x$. (See note in original solution about target value.)
**Solution Steps**

1. From the working: $f(1) = \sqrt{4} + 1 = 3$. So effectively $f^{-1}(3) = 1$ is used.
2. $f'(x) = \frac{1}{2\sqrt{x+3}} + 1$.
3. $f'(1) = \frac{1}{2\sqrt{4}} + 1 = \frac{1}{4} + 1 = \frac{5}{4}$.
4. $(f^{-1})'(3) = \dfrac{1}{f'(1)} = \dfrac{4}{5}$.

$$(f^{-1})'(3) = \frac{4}{5}$$

—

**Question 20:** $f(x) = 3\ln(x) + 2x$. Find $(f^{-1})'(2)$.

**Solution Steps**

1. Find $x_0$: $3\ln(x_0) + 2x_0 = 2$. Testing $x_0 = 1$: $3\ln(1) + 2(1) = 0 + 2 = 2$. So $f^{-1}(2) = 1$.
2. Find $f'(x)$: $f'(x) = \frac{3}{x} + 2$.
3. Evaluate $f'(x_0)$: $f'(1) = \frac{3}{1} + 2 = 5$.
4. Apply the theorem: $(f^{-1})'(2) = \frac{1}{f'(1)}$.

$$(f^{-1})'(2) = \frac{1}{5}$$

## Higher Derivatives and Linear Approximations

**Question 1:** What notation is used for the second derivative of $y = f(x)$ in Leibniz notation?
**Solution:**

$$\frac{d^2y}{dx^2}.$$

—
**Question 2:** If $s(t)$ represents position, what is $s''(t)$?
**Solution:** $s''(t)$ represents the acceleration.
—
**Question 3:** Find $f''(x)$ for $f(x) = x^3 - 6x^2 + 5x + 3$.
**Solution:**

$$f'(x) = 3x^2 - 12x + 5, \quad f''(x) = 6x - 12.$$

—
**Question 4:** Find $f''(x)$ for $f(x) = x \cos x$.
**Solution:**

$$f'(x) = \cos x - x \sin x,$$
$$f''(x) = -\sin x - (\sin x + x \cos x) = -2 \sin x - x \cos x.$$

—
**Question 5:** What is the physical name for $s'''(t)$ where $s(t)$ is position?
**Solution:** The third derivative of position is called the jerk.
—
**Question 6:** What is the $n$-th derivative $f^{(n)}(x)$ for $f(x) = e^{2x}$?
**Solution:**

$$f^{(n)}(x) = 2^n e^{2x}.$$

—
**Question 7:** Find the second derivative $\frac{d^2y}{dx^2}$ implicitly for $x^4 + y^4 = 16$.
**Solution:** First derivative:

$$4x^3 + 4y^3 \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x^3}{y^3}.$$

Differentiating again and simplifying gives

$$\frac{d^2y}{dx^2} = -\frac{48x^2}{y^7}.$$

—
**Question 8:** What is the 27th derivative of $f(x) = \cos x$?
**Solution:** Derivatives cycle every 4 steps.

$$27 \equiv 3 \pmod 4 \Rightarrow f^{(27)}(x) = f^{(3)}(x) = \sin x.$$

—
**Question 9:** State the formula for the Linearization $L(x)$ of $f(x)$ at $x = a$.
**Solution:**

$$L(x) = f(a) + f'(a)(x - a).$$

—
**Question 10:** Define the differential $dy$ for $y = f(x)$.
**Solution:**

$$dy = f'(x)\, dx.$$

—
**Question 11:** Find the differential $dy$ for $y = x^3$ when $x = 1$ and $dx = 0.01$.
**Solution:**

$$f'(x) = 3x^2 \Rightarrow f'(1) = 3, \quad dy = 3 \cdot 0.01 = 0.03.$$

—
**Question 12:** Find the linear approximation $L(x)$ for $f(x) = e^x$ at $a = 0$.

**Solution:**

$$f(0) = 1, \quad f'(0) = 1 \Rightarrow L(x) = 1 + x.$$

—

**Question 13:** Find the linear approximation $L(x)$ for $f(x) = \sin x$ at $a = 0$.
**Solution:**

$$f(0) = 0, \quad f'(0) = 1 \Rightarrow L(x) = x.$$

—

**Question 14:** Find $f'(x)$ for $f(x) = x^{-1}$.
**Solution:**

$$f'(x) = -\frac{1}{x^2}.$$

—

**Question 15:** Explain the relationship $\Delta y \approx dy$ for small $dx$.
**Solution:** $\Delta y$ is the actual change in $f$ over an interval, while $dy$ is the change predicted by the tangent line. For small $dx$, these are approximately equal: $\Delta y \approx dy$.

—

**Question 16:** Find the third derivative $f'''(x)$ for $f(x) = x^5 + 6x^2 - 7x$.
**Solution:**

$$f'(x) = 5x^4 + 12x - 7, \quad f''(x) = 20x^3 + 12, \quad f'''(x) = 60x^2.$$

—

**Question 17:** Find the second derivative $f''(t)$ for $f(t) = t^8 - 7t^6 + 2t^4$.
**Solution:**

$$f'(t) = 8t^7 - 42t^5 + 8t^3, \quad f''(t) = 56t^6 - 210t^4 + 24t^2.$$

—

**Question 18:** Find the linearization $L(x)$ for $f(x) = \sqrt{10 - 3x}$ at $a = 2$.
**Solution:**

$$f(2) = 2, \quad f'(x) = -\frac{3}{2\sqrt{10 - 3x}} \Rightarrow f'(2) = -\frac{3}{4}.$$

$$L(x) = 2 - \frac{3}{4}(x - 2) = \frac{7}{2} - \frac{3}{4}x.$$

—

**Question 19:** Find the linearization $L(x)$ for $f(x) = \ln(x + 3)$ at $a = -2$.
**Solution:**

$$f(-2) = 0, \quad f'(x) = \frac{1}{x + 3} \Rightarrow f'(-2) = 1.$$

$$L(x) = 0 + 1(x - (-2)) = x + 2.$$

—

**Question 20:** Calculate the differential $dy$ for $y = \dfrac{x^2 + 2}{x^4}$ when $x = 1$ and $dx = 0.01$.
**Solution:**

$$y = x^{-2} + 2x^{-4}, \quad y' = -2x^{-3} - 8x^{-5}.$$

At $x = 1$, $y' = -10$, so

$$dy = -10 \cdot 0.01 = -0.1.$$